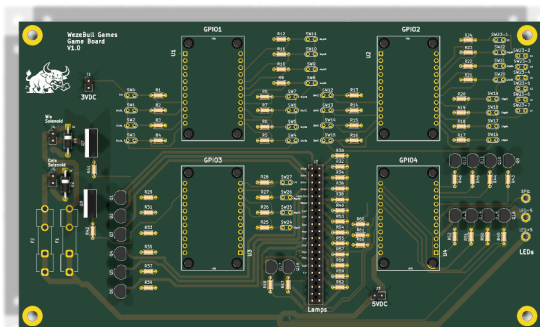




ASSEMBLY GUIDE

SKILLGAME

A TRIBUTE TO THE GAME OF SKILL



ALL HUFF • NO BULL

WezeBull Games · build it once, build it right

Parts list

EVERYTHING YOU NEED BEFORE YOU START

QTY	PART	WHAT IT'S FOR
THE BOARD		
4	Adafruit FT232H breakout	USB→GPIO controllers (U1–U4)
16	PN2222A transistor (TO-92)	lamp / reel drivers
2	TIP120 Darlington (TO-220)	solenoid drivers (Q7/Q8)
2	1N4004 diode	solenoid flyback (D1/D2)
28	10 kΩ resistor	switch pull-downs
16	1 kΩ resistor	transistor base
16	0 Ω jumper (or solder bridge)	lamp links R30–R62 — the #555 bulb self-limits
28	100 nF ceramic cap (0805 SMD)	C1–C28 · switch-input debounce/noise (back of board)
2	2.2 kΩ resistor	TIP120 base
3	5×20 mm fuses · 4 Keystone clips + 1 inline holder	F1/F2 T2A (solenoids) · F3 T3.5A inline (5 V lead to J3)
1	2×20 pin header (J2)	lamp / output connector
4	2-pin headers (J1/J3/J4/J5)	power in + solenoid out
~34	Arcade coin switch (85 mm wire actuator)	28 inputs (S23 = 7 gobble) · coin trips the wire in the slot
1	WS2812b LED strip, 150 px	attract / distract shows
20	#555 wedge LED bulb (6.3 V, non-polar)	16 lamps + a 2nd bulb on 100/200/300/400 (double bright)
20	#555 / T10 wedge socket (twist-lock)	one per bulb; solder its leads to the J2 lamp lines
2	SparkFun Solenoid - 5V (Small)	coin-lock + winner-lock
THE SYSTEM		
1	PELADN Ryzen 5 5600H mini-PC (16 GB / 512 GB)	runs SkillGame
1	QQU 15.6" portable touchscreen (with stand)	the display · 1080P IPS/HDR touch
1	4-port USB 2.0 hub	one port per FT232H
1	3.3 V wall adapter	switch / GPIO logic (J1)

TOOLS

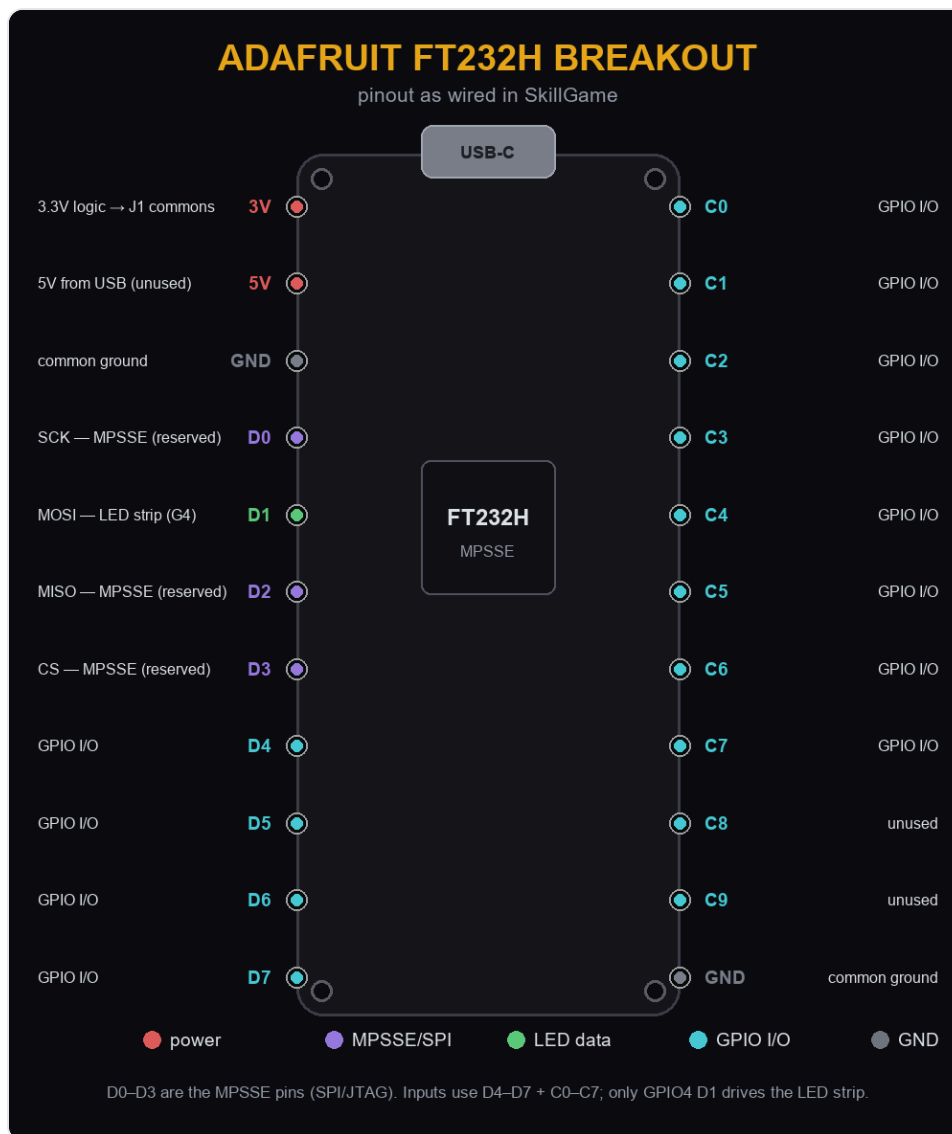
- Soldering iron + solder
- Flush cutters
- Small screwdriver
- Multimeter
- Wire + crimp/ferrules

Before you power up: the 3.3 V adapter goes to J1 (switch commons) and the 5 V to J3 / LED $\pm$. Never feed 5 V into J1 — the FT232H inputs are not 5 V-tolerant.

1

Meet the four boards

Adafruit FT232H — one per GPIO bank



- Four Adafruit FT232H breakouts, each in MPSSE mode over USB-C — the app talks to all four through the USB hub.
- **GPIO1 / GPIO2** read switches; **GPIO3** reads the top rows + tilt and drives the 100–400 reels, Winner, Game-Over and the two coil lines; **GPIO4** drives the 10–90 + Tilt lamps and clocks the LED strip.
- Pins **D0–D3** are reserved (SPI/JTAG) — all switch/lamp I/O lives on **D4–D7** and **C0–C7**. Only GPIO4's D1 is used, for the LED strip.
- Program each board with a unique FTDI serial (GPIO1–GPIO4) so the software binds the right board to the right job no matter which USB port it lands on.

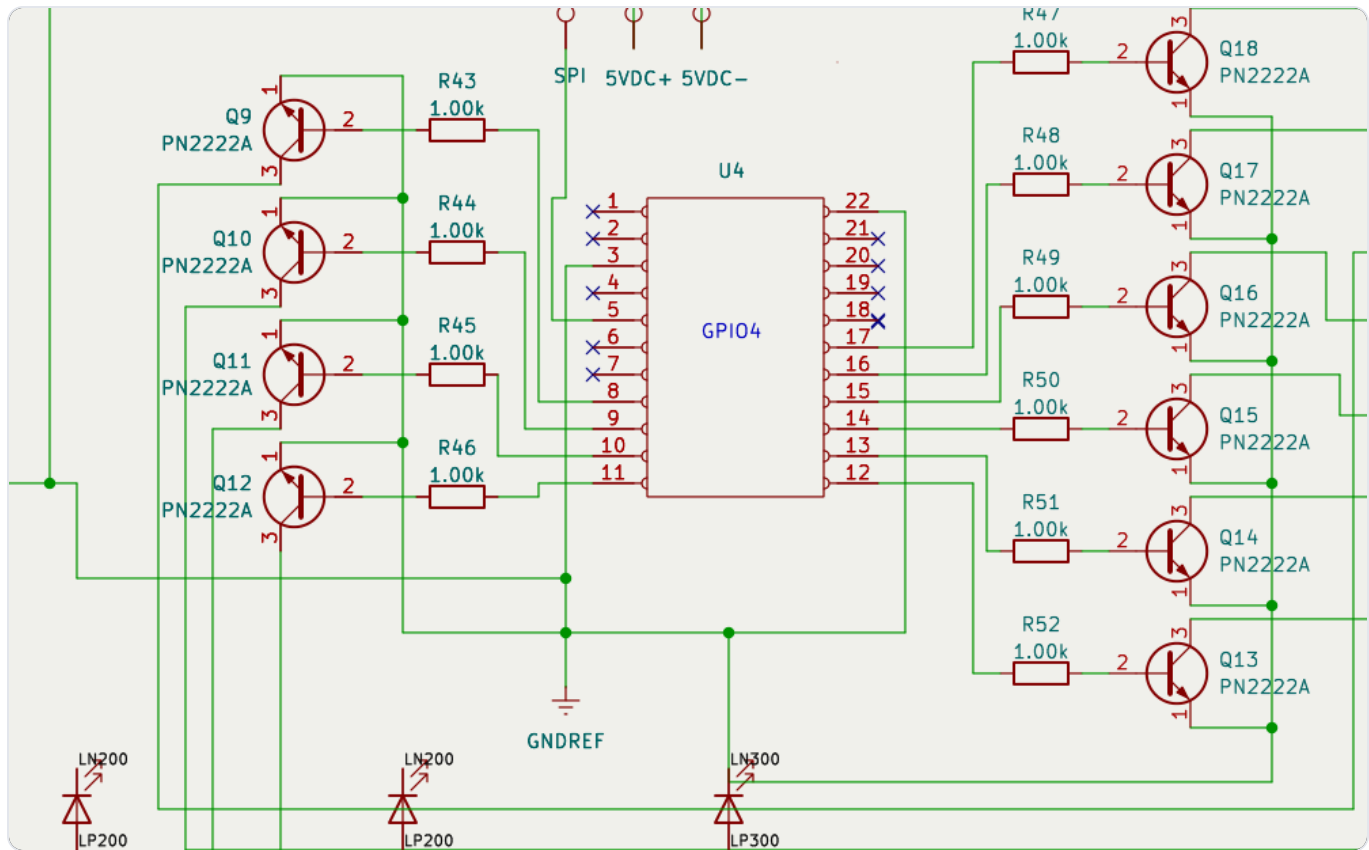
- The FT232H's own 3 V / 5 V pins stay unconnected — the boards are USB-powered.

SkillGame Assembly Guide

2

Solder the drivers

resistors, then transistors — lowest parts first

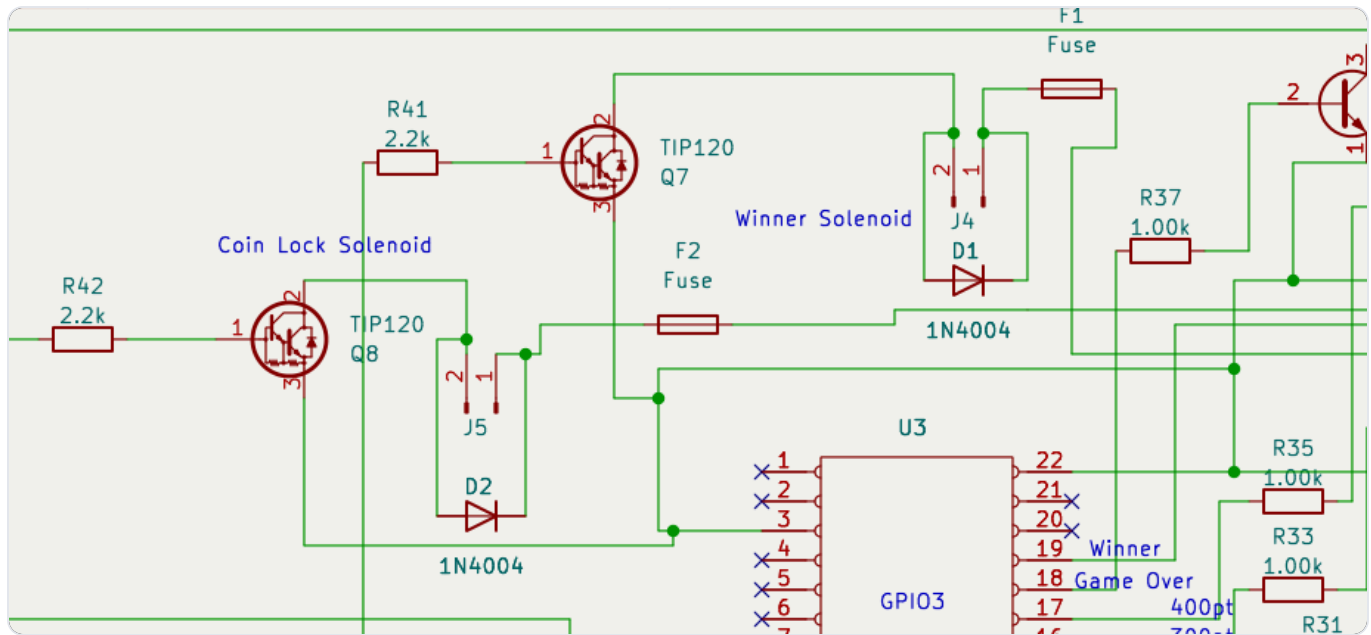


- Resistors first (they lie flat): 28× 10 kΩ pull-downs, 16× 1 kΩ bases, 2× 2.2 kΩ TIP120 bases. The 16 lamp positions (R30–R62) are **0 Ω links** — a jumper wire or solder bridge, since the #555 bulbs self-limit.
- Then the 16 PN2222A transistors (TO-92) — mind the flat/rounded face matches the silk.
- Keep leads short; flush-cut the backs.

3

Solenoid drivers + fuses

TIP120s, flyback diodes, F1/F2



⚠ **TIP120 orientation:** collector goes to the coil, emitter to ground. If you are using an **older board revision**, cross the collector (middle/tab) and emitter (right) leads on Q7 & Q8 — a straight insert wires them backwards and the coils won't switch. The current KiCad files already have this fixed.

- Fit the two TIP120s (TO-220) and the 1N4004 flyback diodes — band matches the silk.
- Install the Keystone 3517 clips and drop in the fuses (see next page).

3b

Fuse the solenoids

F1 / F2 — one per coil, on the board

FUSES

F1 · F2 solenoids · F3 main 5 V rail

Ref	Protects	In line with	Rating	Part number	Size
F1	Winner-Lock solenoid	+5 V feed to J4	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F2	Coin-Lock solenoid	+5 V feed to J5	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F3	Main 5 V rail (inline)	5 V supply lead into J3	T3.5A slow-blow 250 V	Bussmann GDC-3.5A	5 x 20 mm

Holders: F1/F2 use Keystone 3517 board clips (2 each); F3 uses an inline 5 x 20 mm holder in the 5 V supply lead.

F1/F2 protect the two small 5 V solenoids (SparkFun 'Solenoid - 5V Small') - T2A slow-blow covers their inrush.

F3 guards the whole shared 5 V rail (lamps + LED strip + coils) right where the 5 V adapter feeds J3.

Slow-blow (T) rides out coil / LED inrush; any 250 V 5 x 20 mm part clears 5 V fine.

- Solder the four **Keystone 3517** clips — two per fuse — at the F1 and F2 positions. They only go one way; the fuse snaps in across the pair.
- Drop in the fuses: **F1** guards the Winner-Lock coil (J4), **F2** the Coin-Lock coil (J5). Both are **T2A slow-blow, 5×20 mm** (Littelfuse 0218002.MXP).
- Slow-blow (T) rides out the coil's turn-on surge; it opens only on a real short. Any 250 V 5×20 mm part is fine on the 5 V rail.

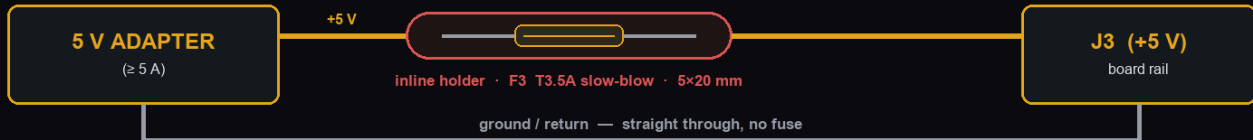
3c

The inline main fuse

F3 — one fuse that guards the whole 5 V rail

THE INLINE MAIN FUSE · F3

splices into the +5 V wire feeding J3 — guards the whole lamp / LED / coil rail



- Cut the +5 V wire from the adapter and put the inline holder in the break — screw-terminal ends, 18 AWG.
- F3 = T3.5A slow-blow. It rides out lamp / LED / coil inrush but opens on a hard short before the traces do.
- F1 / F2 (T2A, on the board in Keystone clips) still protect each solenoid on top of F3.
- Part: uxcell inline screw-type 5×20 mm holder, 18 AWG (Amazon B07SM5KYZ7).

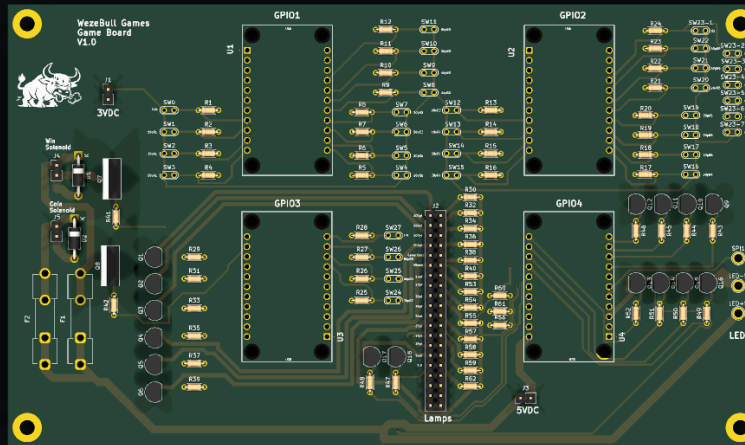
- F1/F2 only protect the coils. **F3 protects everything on 5 V** — lamps, LED strip and coils — right where the adapter feeds the board.
- It isn't on the PCB: cut the **+5 V** wire coming from the adapter and splice the **inline holder** (uxcell 18 AWG, screw-terminal) into the break. Leave the ground/return wire whole.
- Fit a **T3.5A slow-blow, 5×20 mm** fuse (Bussmann GDC-3.5A). Sized above the full lamp + strip + coil draw, below the adapter's limit.

Rule of thumb: $F3 \approx \text{your steady current} + \text{inrush headroom}$, and always below what the wiring can carry. 3.5 A suits a ≥ 5 A adapter here.

4

Headers & sockets

J1–J5, the J2 lamp header, and the 4 FT232H sockets

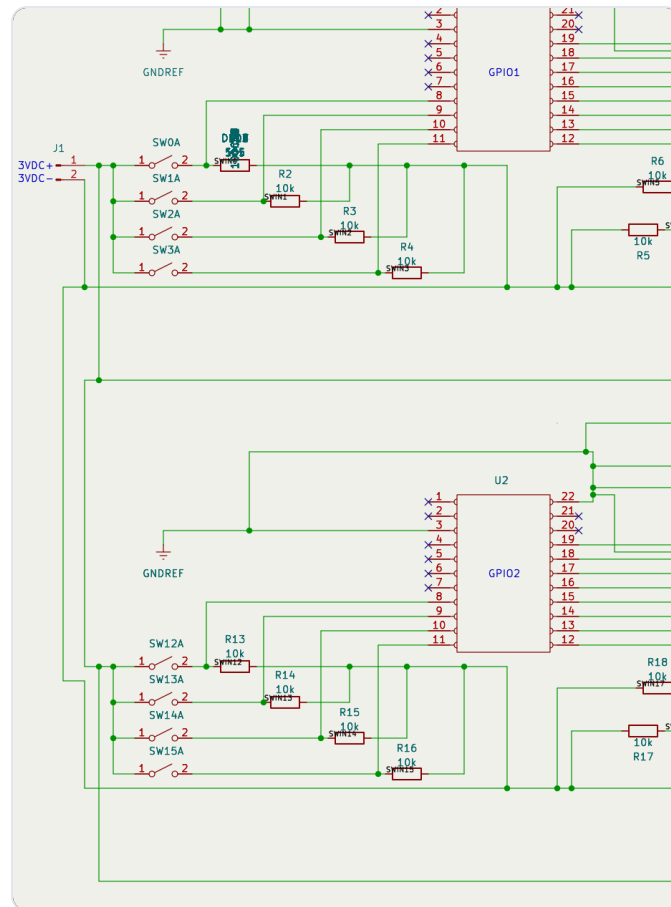


- Solder the **2×20 header (J2)** — the lamp/output connector — and the four **2-pin headers**: J1 (3.3 V in), J3 (5 V in), J4 (Winner-Lock), J5 (Coin-Lock).
- Tack one pin first, check the header sits flat and square, then solder the rest — a skewed 40-pin header is miserable to fix.
- Fit the four FT232H sockets with even spacing so the boards sit flat and don't touch. The footprints already have clearance; don't skew them.
- Note which socket is GPIO1–GPIO4 (silk) — it must match the serial you programmed into each board.

5

Wire the switches

28 inputs → J1 commons, pulled to ground



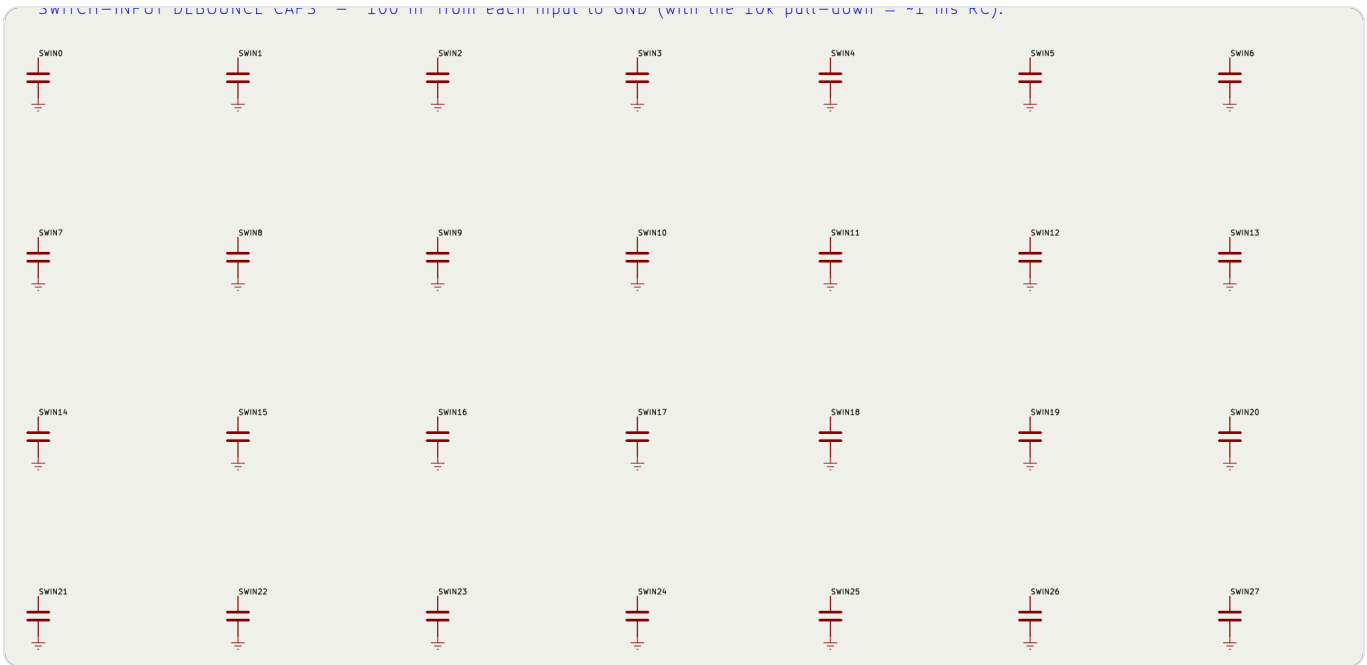
- **28 inputs** across GPIO1–3 (D4–D7, C0–C7). One side of every playfield switch ties to the shared **J1 3.3 V common**; the other side goes to that switch's FT232H input pin.
- An on-board **10 kΩ pull-down** holds each input LOW; closing the switch pulls it to 3.3 V (a real HIGH) and the software reads the hit.
- **Keep the common at 3.3 V — never 5 V.** The FT232H inputs are **not** 5 V-tolerant; 5 V on an input can kill the chip.
- The **7 gobble holes** all report as one input (S23): wire their switches **in parallel** so any of them trips it.
- Run it as a harness (see the last pages): one 3.3 V bus wire daisy-chained to every switch, plus one signal wire per switch back to the board.

5b

Fit the input caps

28× 100 nF on the board's back — debounce & noise

SWITCH-INPUT DEBOUNCE CAPS = 100 nF from each input to GND (with the 10k pull-down = ~1 ms RC).

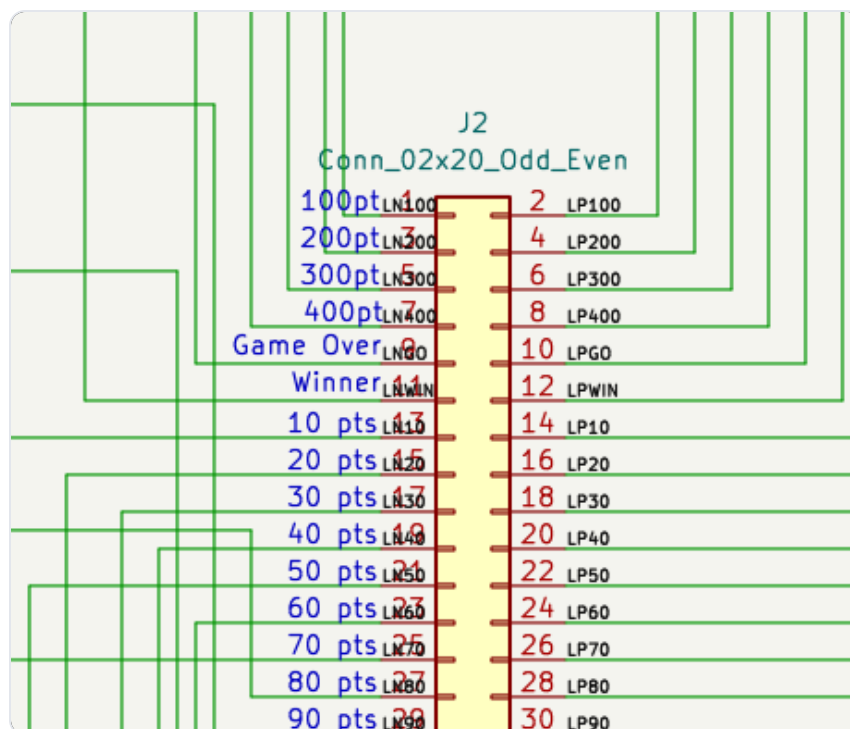


- Each switch input gets a **100 nF (0.1 μF) ceramic cap to GND** — C1–C28. With the 10 kΩ pull-down it forms a ~1 ms RC filter that kills contact chatter and noise picked up on the long playfield leads.
- They're **SMD 0805 on the back of the board**, one tucked next to each pull-down (GND pad on the ground pour, the other to the input). Solder them last.
- Optional but recommended — the game still works without them (the software also debounces), but they make the long switch runs far more reliable.

6

The score lamps

off-board #555 wedge LED bulbs on J2



- Every score lamp — the tens **10–90**, the reels **100–400**, and **Winner / Game-Over / Tilt** — is a **#555 wedge LED bulb** in a twist-lock socket, off the board.
- Each bulb plugs across one **J2 pin pair**. The **odd** pin (left, labelled with the lamp name) is switched by the transistor; the **even** pin (right) is +5 V through that lamp's 0 Ω link. The labels on J2 tell you which pair is which lamp.
- Solder the socket's two leads to that pin pair. The #555 bulbs are **non-polar**, so which lead goes to which pin doesn't matter.

Wire it as a harness (see the last pages): a 2×20 plug on J2, fanning out to all the sockets, is far tidier than 40 loose wires.

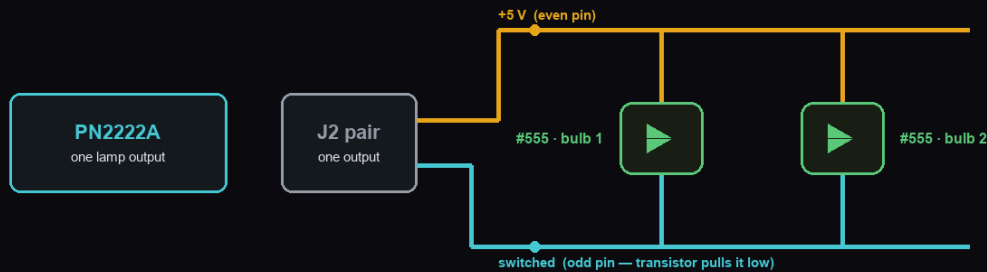
6b

Double up the big holes

100 / 200 / 300 / 400 — two bulbs each

DOUBLE-BRIGHT HOLES · 100 / 200 / 300 / 400

two #555 bulbs in parallel on one J2 output — each runs at full brightness



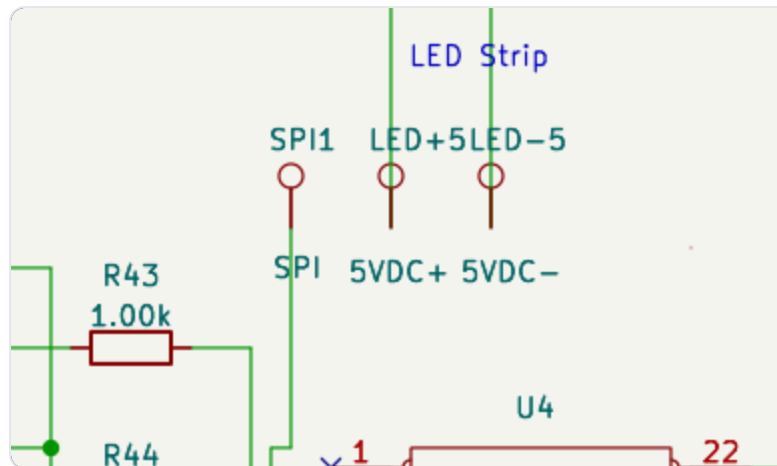
- Both bulbs' + leads land on the even (+5 V) pin; both other leads on the odd (switched) pin. Straight parallel.
- No second resistor needed — each bulb self-limits, so both get the full 5 V and full brightness.
- #555 bulbs are non-polar, so socket orientation doesn't matter. Only 100 / 200 / 300 / 400 get two bulbs.

- The four high-score holes each drive **two** #555 bulbs for extra punch. Wire the second bulb **in parallel** on the same J2 pair as the first.
- Both bulbs' + leads land on the even (+5 V) pin; both other leads on the odd (switched) pin.
- No extra resistor — each bulb self-limits, so both run at full brightness. The transistor easily drives two.

6c

Coils & the LED strip

solenoids off J4 / J5, strip off SPI

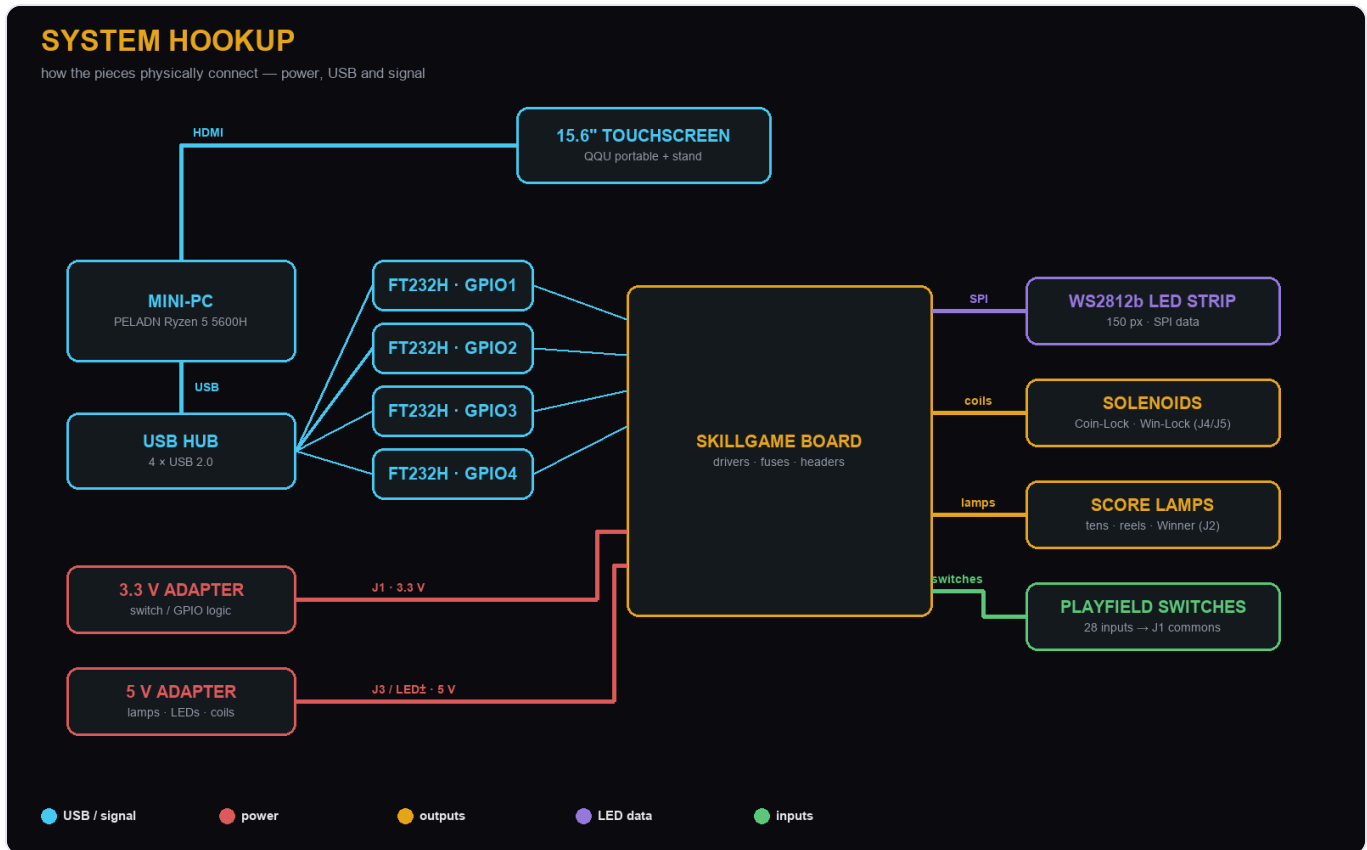


- **Coin-Lock** → **J5**, **Winner-Lock** → **J4**. These are the fused (F1/F2), TIP120-driven 5 V coils — keep the flyback diode across each coil.
- LED strip **data** → **GPIO4's D1** (SPI MOSI). Data is one wire plus a common ground.
- Power the strip from its **own 5 V feed** (LED+5 / LED-5), **not** J3, so its heavy draw can't sag the lamp/logic rail. Tie all grounds common.

7

Hook up the system

PC · hub · monitor · power



- **Mini-PC** → **HDMI** → **monitor** (the QQU 15.6" touch panel), and its USB-C touch lead back to the PC.
- **Mini-PC** → **USB hub** → **the four FT232H boards** — one hub port per board.
- **3.3 V adapter** → **J1** (switch commons). **5 V adapter** → **J3** (through the F3 inline fuse) and to the LED strip's own **LED± feed**.
- **Tie every ground together** — the 3.3 V return, the 5 V return, the LED strip and the boards all share one common ground, or nothing reads right.
- Double-check J1 is 3.3 V and J3 is 5 V before powering on. They are not interchangeable.

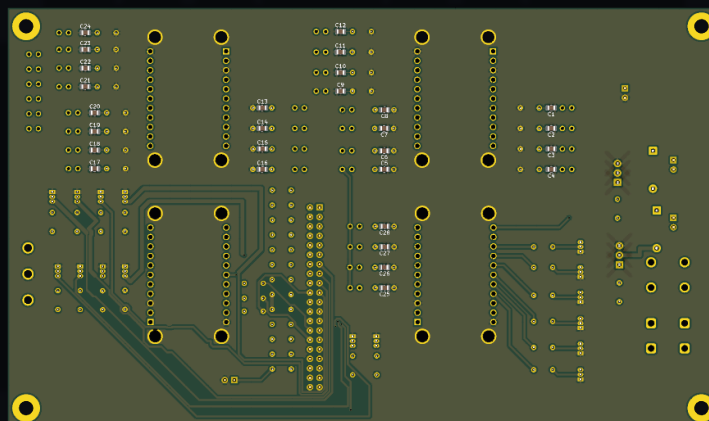
8

Power on & self-test

let the software check your wiring

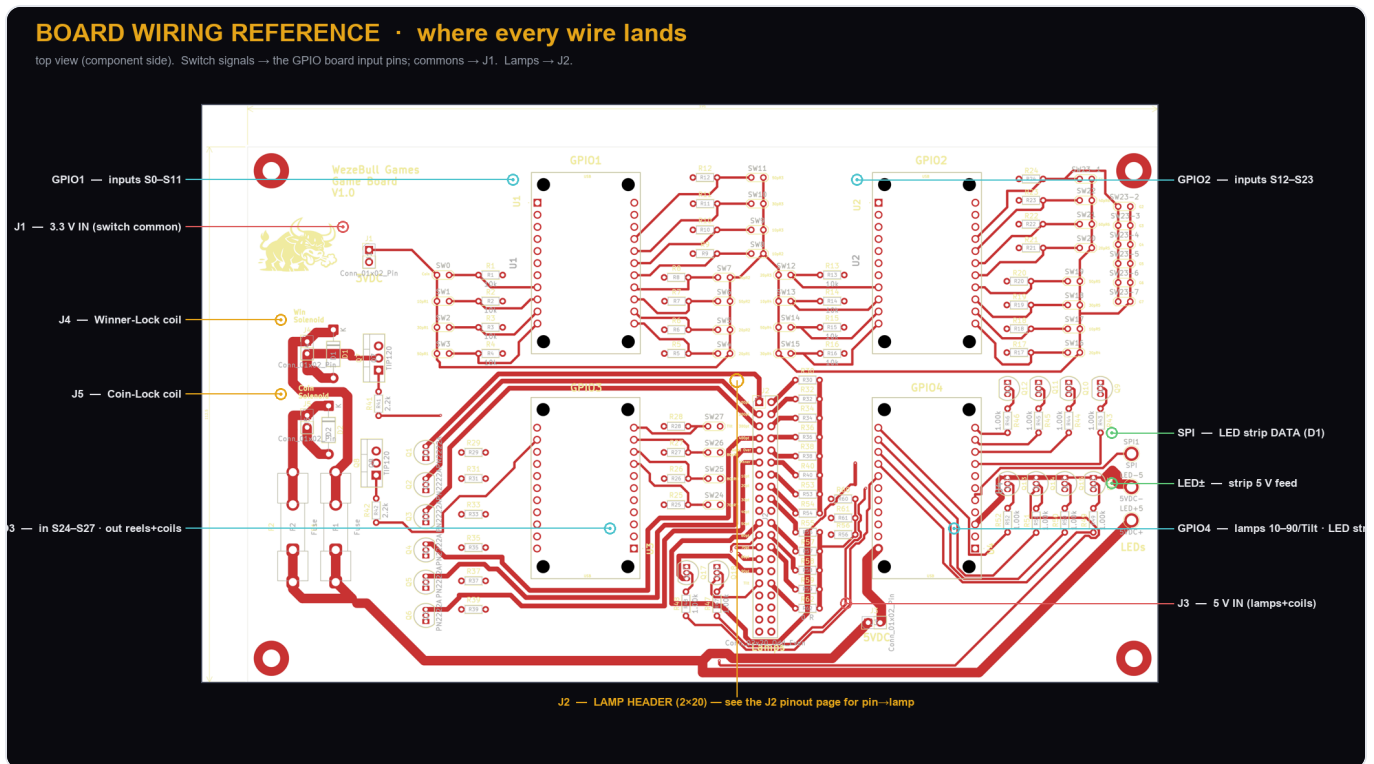
- Launch SkillGame, open **Diagnostics**, and run **Self-Test** — it lights every lamp, pulses the coils and sweeps the LED strip.
- Use the **Switch Monitor** to press each playfield switch and confirm it registers (and clears).
- Any input stuck HIGH shows as a fault ring — check that switch's wiring and pull-down.
- Green badge on Diagnostics = all boards present and healthy. You're done. **ALL HUFF • NO BULL.**

The board back is on the next page for reference, followed by the full schematic.



Wiring reference

WHERE EVERY WIRE LANDS ON THE BOARD



- Every connector and GPIO socket is called out above. **J1** = 3.3 V in (switch common), **J2** = the 2×20 lamp header, **J3** = 5 V in, **J4/J5** = the two coils, **SPI + LED±** = the LED strip.
- Switch signals land on the **GPIO board input pins** (next page); lamp wires land on **J2** (pin→lamp map is on the J2 page).



Switch-input pin map

which pin every switch wire goes to

SWITCH INPUT WIRING

each coin switch: COM → the shared 3.3 V bus (from J1) · NO → the FT232H pin below (on that GPIO board)

GPIO 1 socket U1			GPIO 2 socket U2			GPIO 3 socket U3		
S0	D4	Coin Up	S12	D4	20 pts	S24	D4	70 pts
S1	D5	10 pts	S13	D5	10 pts	S25	D5	80 pts WINNER
S2	D6	30 pts	S14	D6	50 pts	S26	D6	80 pts WINNER
S3	D7	50 pts	S15	D7	30 pts	S27	D7	TILT
S4	C0	20 pts	S16	C0	20 pts			
S5	C1	20 pts	S17	C1	10 pts			
S6	C2	50 pts	S18	C2	30 pts			
S7	C3	30 pts	S19	C3	50 pts			
S8	C4	10 pts	S20	C4	20 pts			
S9	C5	10 pts	S21	C5	60 pts			
S10	C6	30 pts	S22	C6	40 pts			
S11	C7	50 pts	S23	C7	GOBBLE x7			

HOW TO WIRE A SWITCH

- COM terminal → the 3.3 V common bus (one wire daisy-chained to every switch, fed by J1).
- NO terminal → the pin shown (D4-D7 / C0-C7 on that GPIO board's FT232H).
- Leave NC unused.
- The 7 GOBBLE switches all land on ONE pin (S23 / GPIO2 C7) — wire their NO's in parallel.

Each coin switch is a plain SPDT: **COM** → **the 3.3 V common bus** (fed by J1), **NO** → **the FT232H pin shown**. NC is unused. The 7 gobble switches all wire in parallel onto the one S23 pin (GPIO2 · C7).

Wiring harnesses

THE CABLES THAT JOIN THE PLAYFIELD TO THE BOARD

- The board only holds the transistors, resistors and connectors. The **lamps** (backglass) and **switches** (playfield) live off-board, so two harnesses connect them:
- **Lamp harness** — a 2×20 plug on J2 fans out to the 20 #555 wedge sockets.
- **Switch harness** — one 3.3 V common bus to every switch, plus a signal wire from each switch back to the board.
- Plus the **inline fuse cable** (F3) spliced into the 5 V feed.
- Each cable is shown three ways on the next pages: flat with every solder point, a 3D view, and **where every point lands on the real game.**



Lamp harness — wiring

J2 plug → 20 #555 sockets, every solder point

LAMP HARNESS

one 2×20 plug on J2 → 20 #555 wedge sockets · gold = +5 V (even pin), cyan = switched (odd pin)



- Solder each socket's two leads to its J2 pin pair (dots). 100 / 200 / 300 / 400 carry a second socket on the same pair.
- GROUND closes ON THE BOARD — the driver transistor's emitter to GND. Only 2 leads run to each socket, no ground wire.
- Non-polar #555 bulbs — either lead to either pin. Keep +5 (gold) and switched (cyan) consistent so the harness stays readable.

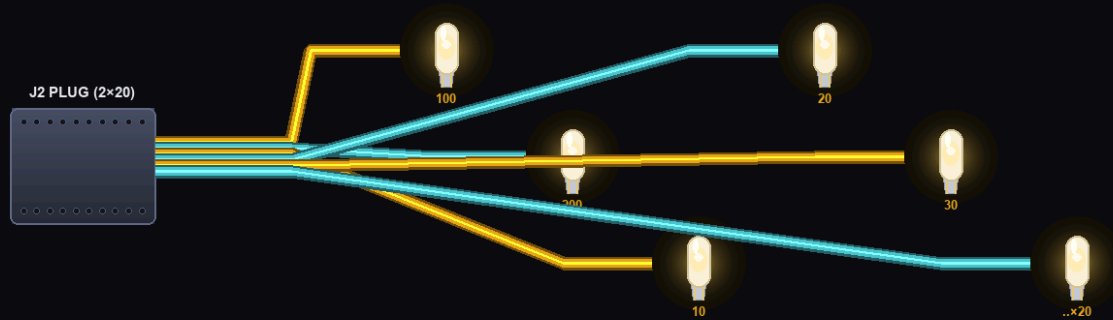


Lamp harness — 3D & where it lands

the physical cable, and each bulb's spot on the backglass

LAMP HARNESS · 3D

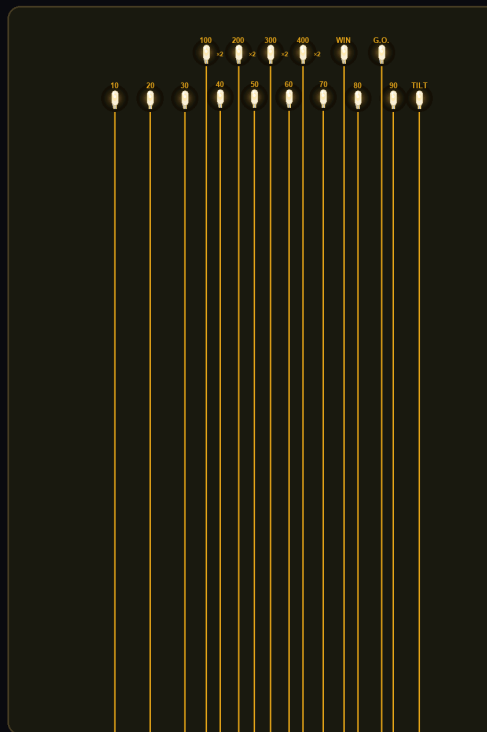
a 2×20 plug on J2 fans out to 20 #555 wedge sockets (6 shown)



Route as a bundle from the plug, break out to each socket. Doubled holes (100–400) carry a second socket on the same wire.

LAMP HARNESS · on the backglass

every #555 bulb socket in its real spot — gold = +5 V, green = switched lamp line from J2



CONTROL BOARD · J2 = 2×20 lamp header



Switch harness — wiring

3.3 V bus + one signal per switch, every solder point

SWITCH HARNESS

28 inputs · one 3.3 V common bus (from J1) to every switch, one signal wire per switch back to the board



- Every switch = just 2 leads: one to the shared 3.3 V bus (red), one back to its board input (cyan). Solder at every dot.
- GROUND closes ON THE BOARD — a 10 kΩ pull-down on each input to GND. No ground wire runs out to the switch.
- The 7 gobble-hole switches wire in PARALLEL onto one input (S23) — any one trips it. 3.3 V only, never 5 V.
- Optional debounce: a 100 nF cap from each input to GND (on the board) filters contact chatter + noise on the long leads.

Which switch? Use an **arcade coin switch with an 85 mm straight wire actuator** (SPDT, ~\$3.99 ea; ~\$2 in 10-packs on eBay/AliExpress). It's a snap-action microswitch with a long thin wire — the wire pokes up through a slot in the playfield and the rolling coin trips it. The long wire gives so much leverage that a light coin trips it easily, and the switch body mounts **hidden under the playfield**. It wires like any SPDT switch (COM + NO), so no board changes. This is the same style Bally used.

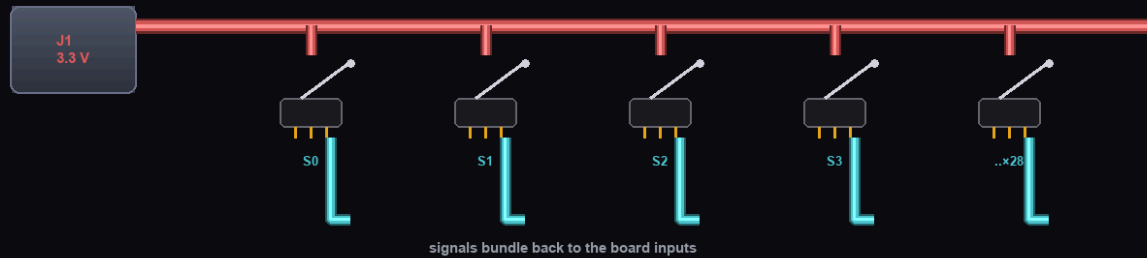


Switch harness — 3D & where it lands

the physical cable, and every switch's spot on the playfield

SWITCH HARNESS · 3D

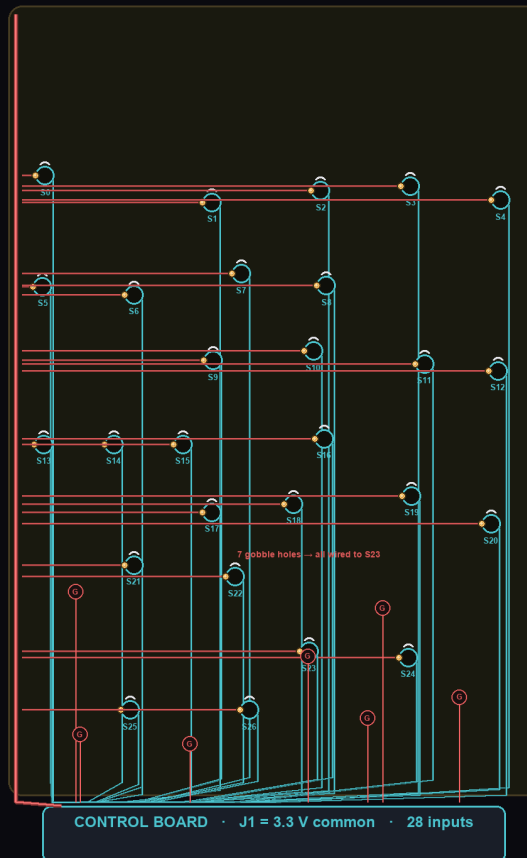
one 3.3 V bus daisy-chained to every switch, a signal wire from each back to the board (5 shown)



The long wire actuator pokes up through the playfield slot; the rolling coin trips it. 3.3 V bus in red, per-switch signal in cyan.

SWITCH HARNESS · on the playfield

every scoring / gobble hole switch in its real spot — red = 3.3 V common bus, cyan = signal to the board





Inline fuse cable — 3D

F3 in the 5 V feed to J3

INLINE FUSE CABLE · F3

splices into the +5 V lead from the adapter to J3



Cut the +5 V wire, screw each end into the inline holder, drop in the T3.5A fuse. 18 AWG (uxcell B07SM5KYZ7).

